

WPL DATA ANALYSIS 2023-2025

End-to-end analysis of Women's Premier League cricket matches — combining SQL, Excel, and Python to uncover team performance, toss impact, and venue trends across the league's first three seasons.

36

MATCHES

03

SEASONS

05

TEAMS

11

SQL QUERIES

10+

VISUALS

PROBLEM STATEMENT

The Women's Premier League generates match-level data every season, but no consolidated view exists of how teams, tosses, venues, and win margins have trended since the league's 2023 launch. This project cleans and analyzes that match data end-to-end to surface those trends and translate them into clear, actionable insights.

TOOLS & APPROACH

PYTHON · JUPYTER

- Data cleaning & type fixes
- Pandas groupby analysis
- Seaborn / Matplotlib charts
- Optional ML win predictor

SQL · SQLITE

- Relational schema design
- Aggregations & joins
- Win % & toss-impact queries
- Season champion lookup

EXCEL DASHBOARD

- Live COUNTIF / SUMPRODUCT formulas
- 5-sheet interactive workbook
- Native bar & pie charts
- Auto-updates from raw data

KEY INSIGHTS

- Toss ≠ Match Outcome** — the toss winner went on to win the match less than half the time, showing on-field performance outweighs toss luck.
- Fielding-First Bias** — teams overwhelmingly chose to field first after winning the toss, mirroring the modern T20 chasing trend.
- Early Team Dominance** — a small set of franchises account for a disproportionate share of total wins across the first three seasons.
- Venue Concentration** — matches are hosted at a limited number of venues, reflecting the league's early-stage scheduling footprint.
- Competitive Margins** — win margins skew moderate rather than one-sided; most matches stay close until the final overs.

Pandas

NumPy

Seaborn

Matplotlib

SQLite

openpyxl

Scikit-learn

Jupyter Notebook

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